

Multiple Choice

1. **Answer:** A
Options: A) `int(x [,base])`; B) `long(x [,base] )`; C) `float(x)`; D) `str(x)`
Note: Correct option: A - `int(x [,base])`
2. **Answer:** C
Options: A) $O(N^{1/2})$; B) $O(N^{1/4})$; C) $O(N^{1/N})$; D) $O(N)$
Note: Correct option: C - $O(N^{1/N})$
3. **Answer:** D
Options: A) 1; B) 6; C) 4; D) 5
Note: Correct option: D - 5
4. **Answer:** B
Options: A) a; B) Chemistry; C) 0; D) 1
Note: Correct option: B - Chemistry
5. **Answer:** B
Options: A) 0; B) 1; C) 3; D) 4
Note: Correct option: B - 1

True / False

6. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'E 7_{16}'.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def divSum(n): | return sum; | def areEquivalent(num 1,num 2): | return divSum(num 1) == divSum(num 2);`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def set_middle_bits(n): | return (n >> 1) ^ 1 | def toggle_middle_bits(n) : | return 1 | return n ^ set_middle_bits(n)`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key